

BCIMS PROJECT REPORT

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BIHAR COVID INFORMATION MANAGEMENT SYSTEM

OVERVIEW

The Bihar Covid Information Management System (BCIMS) is designed to help record and accumulate data of travellers coming to Bihar. The data recorded acts as an aid for policy decisions.

DELIVERED

- **Data Entry System:** Entry of travellers' data at district level
- **Admin Panel with Dashboard:** Admin panel at district level for report generation of travellers and monitoring, answering queries and access to Dashboard
- **BCIMS Dashboard:** Data analytics and report generation at department level. Detailed analytics and queries answered at state level. District wise report generation.
- **Garuda Dashboard:** Dashboard, data analytics and report generation of monitoring of travellers at department level. District wise report generation of monitoring status.
- **Chakshu Mobile App:** Mobile app to track and monitor travellers during their quarantine period. Real time reporting to district and state level. Has a feature of geotagging for authorization of monitoring data.
- **Complaint Management System:** A system to record complaints at RC Bihar, LC Bihar, CM Office Bihar, DMD Bihar and District Control rooms was put in place to handle the requests and complaints that were being made by people especially migrants living outside the home state
- **Password Management System:** A password management system was created for IT admin.

TECH SPECIFICATIONS

Server Spec (Cloud): Ubuntu 16.04.6 (LTS) x64 | 16 vCPUs Dedicated | 32 GB RAM | 160GB SSD

Server Spec (SDC)

Application Server: Ubuntu 16x | 120GB SSD | 64GB RAM | 16 Core CPU

Database Server: Linux RHEL 7x | 120GB SSD | 16GB RAM | 4 Core CPU

Framework: The project is developed on Codeigniter Framework of PHP with use of bootstrap, JavaScript and jQuery.

Database: MySQL Database is used in cloud server at staging and Maria DB Community Edition on SDC.

Frontend: jQuery, jQuery – Mobile, Angular JS, Google Fonts, Bootstrap, Masonry v3, Font-Awesome icons.

Editors for content creation: CK-Editor. (WYSIWYG)

Data Parsing & WebServices: RESTFUL API using JSON

SSL Security: SSL Certificate bought from GoDaddy with 1 year of Validity.

USER ACCESS

User System and Controlled Level of Access(CLA): Five types of user-level access have been created.

A. Super Admin - DMD Level: Gets access to complete data and analysis, Can filter reports District Wise, Travel Date Wise, and Type of Travel wise etc.

B. Admin - District Level Access: Controls everything at their district level, from data entry to reports.

C. Field Exec - Block & Panchayat Level: Has access to the mobile app and collects monitoring data in the field.

D. IT Admin - Manages user passwords

E. Executive - Access to complaint management system to record complaints received at RC Bihar office, LC Bihar office and CM office.

AUTOMATIC BACKUP OF DATA & MAINTENANCE

A database backup system was implemented at the state datacenter server in which an automatic daily backup of the entire database is created at the server and is managed and stored in physical tape by the state data center. Daily, Weekly and Monthly backups are being maintained.

Development Methodology

We have followed Agile development methodology instead of the traditional development models (like waterfall model) as Agile development methodology provides opportunities to assess the direction of a project throughout the development lifecycle and hence fewer major changes and missed bugs in the final project.

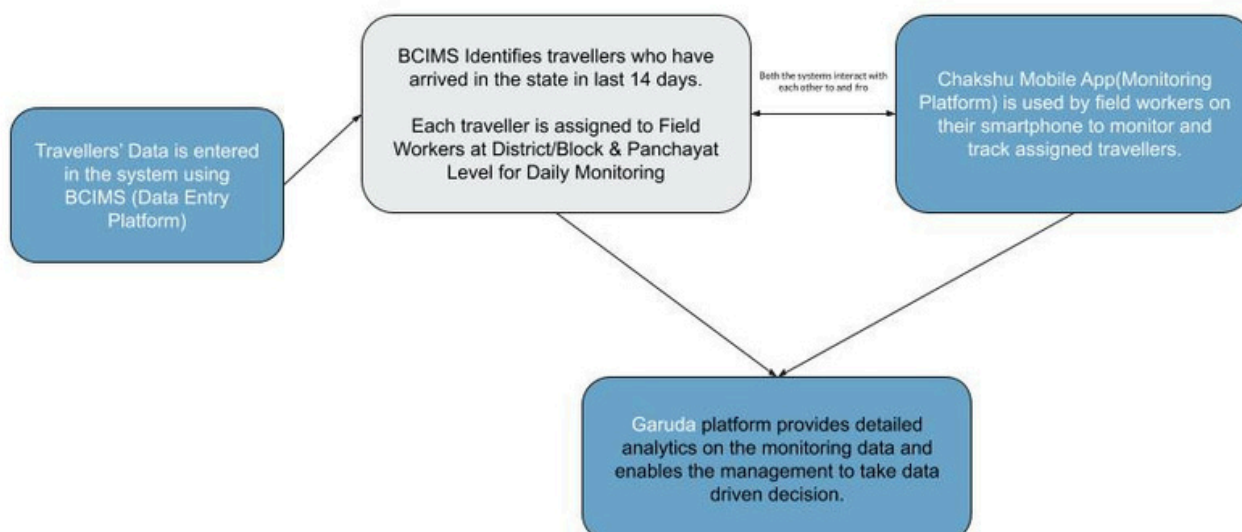
Tool

Project Management: Mantis with Agile Plugin – Mantis, a web-based, tool was used to manage projects and sprints. All project developments, issues, user stories, and requirements can be tracked.

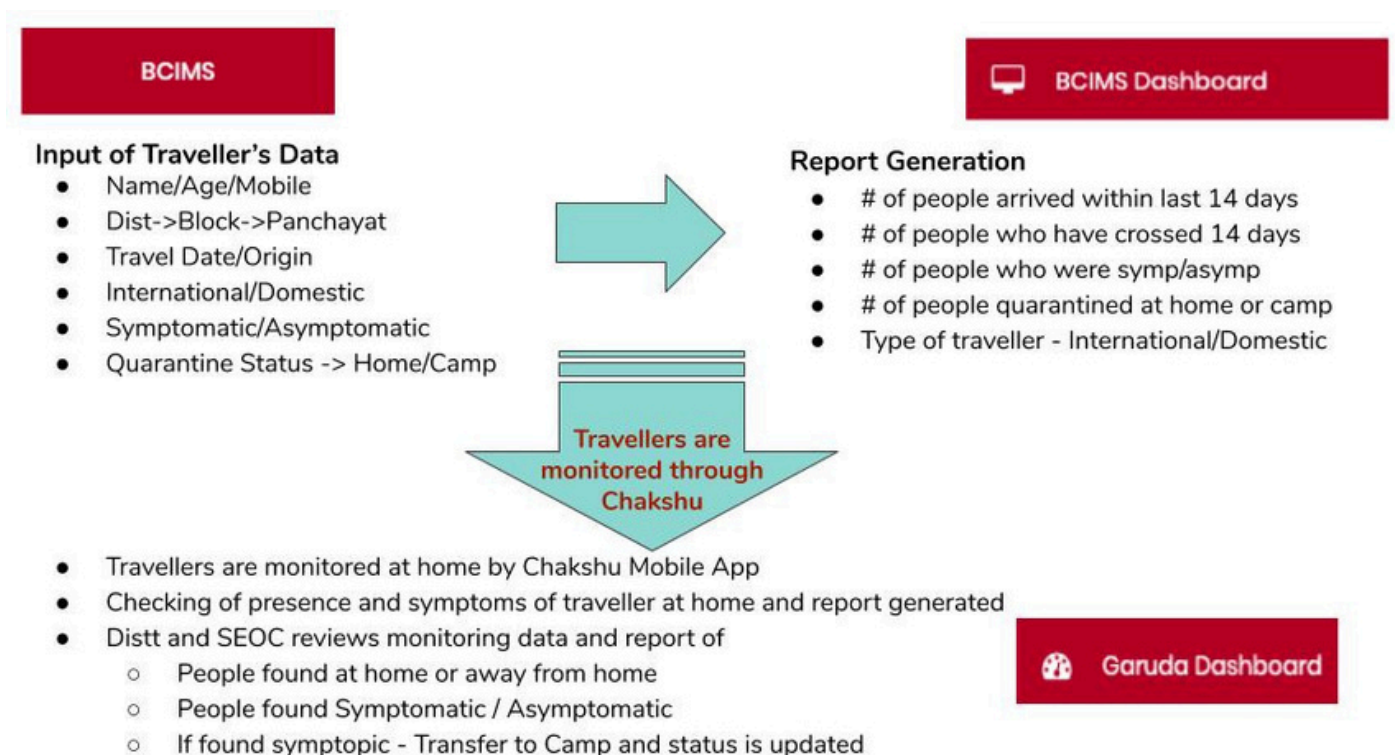
Version Control & Git: Bitbucket – Bitbucket private repository was used for version control and development with 3 branches viz. Dev, Staging, and Production. This helped us in maintaining versions during development, testing, and production and also prevented us from losing any previously developed version/feature.

DIAGRAMS, FLOWS & SCREENSHOTS

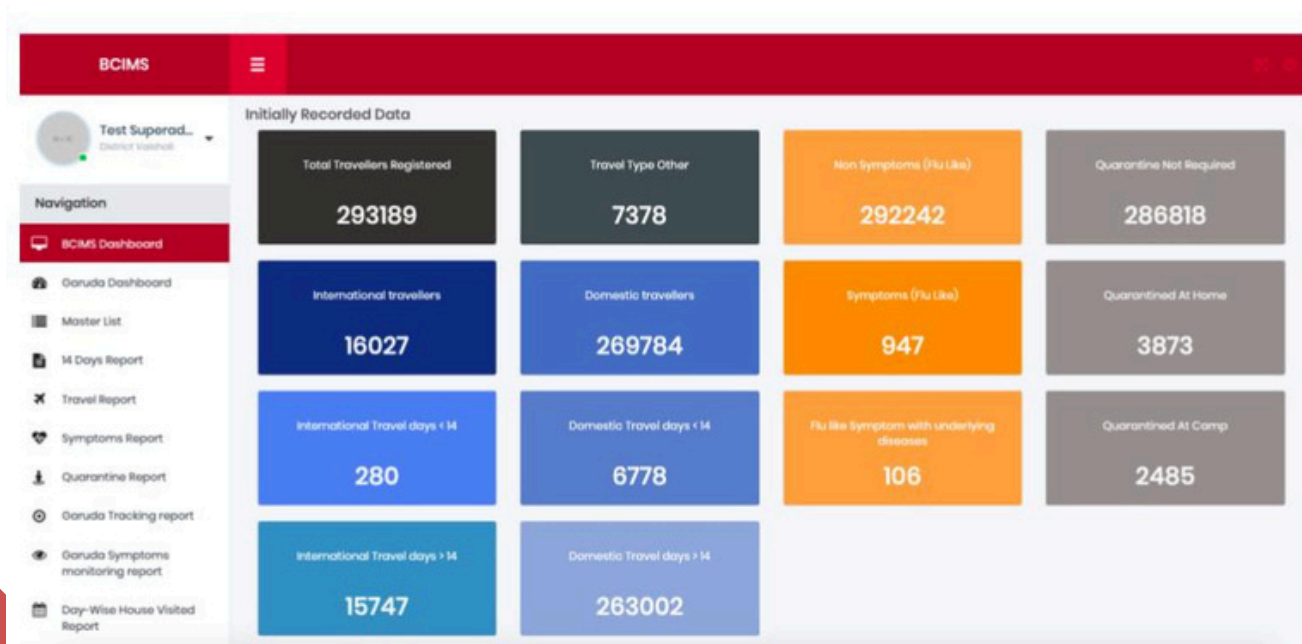
Data Flow Diagram



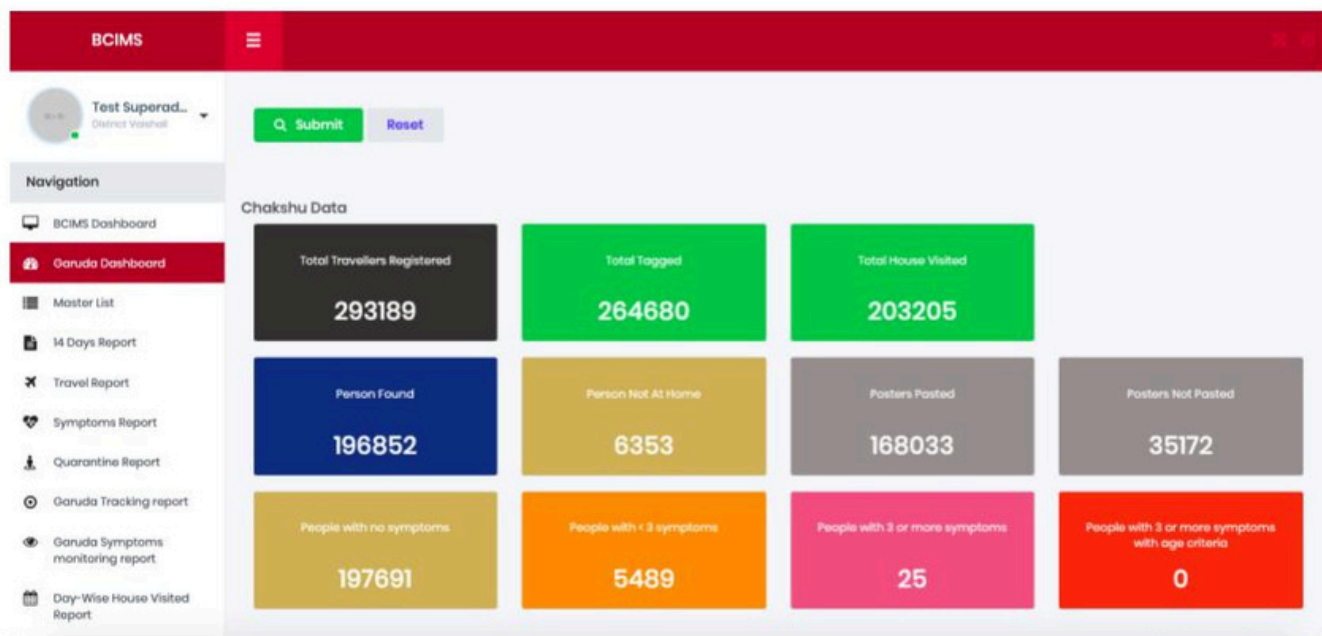
Correlation Between BCIMS, Garuda and Chakshu



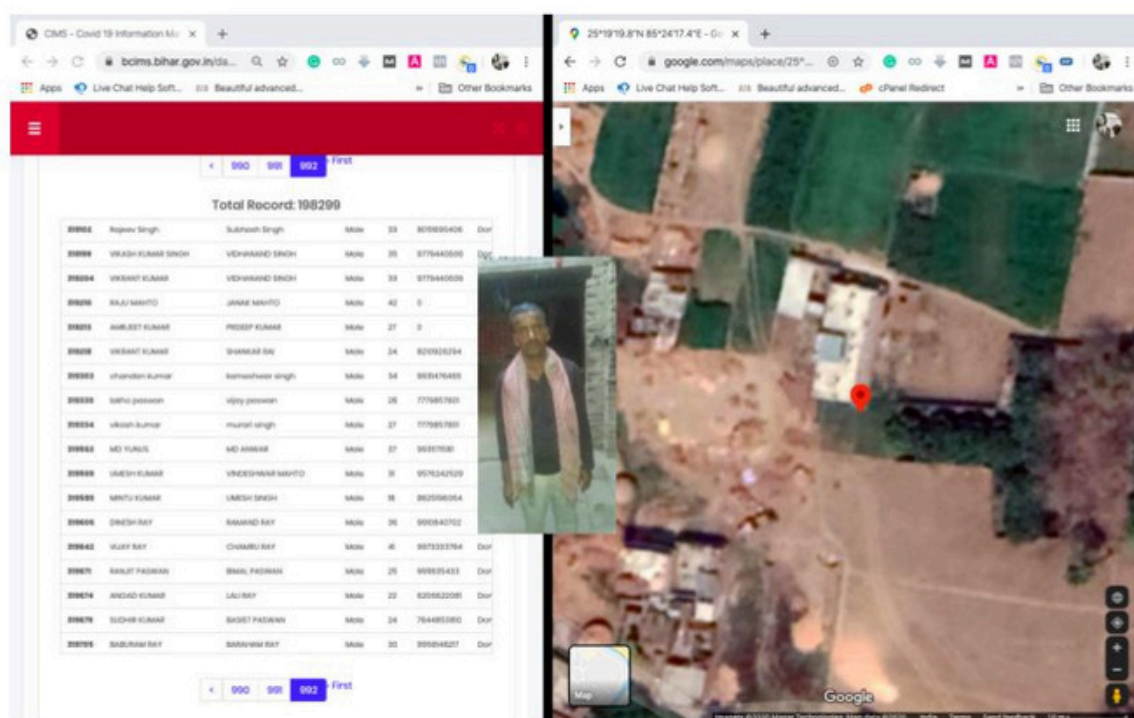
BCIMS Dashboard



GARUDA Dashboard



Geotagged Monitoring Report of an Individual



Data view and Filtering Options at District Level

Travel Data View - Vaishali

(Logged in As Test Superadmin)

Logout
New Traveller's Entry
New Complaint Entry
Go To Dashboard

All Data
14 Days Data
Monitoring Report
Daily Monitoring Report
Assigned Report
Pending Cases

Deleted Records
Closed Cases
Complaint List

Block

All

Panchayat

All

Travel Type

All

Symptom

All

Quarantine

All

Status

All

Search By Unique Id

Search By Name

Search By Mobile No.

Search By Unique Id

Search By Name

Search By Mobile No.

Submit
Reset
Export Excel

Daily Monitoring Report at District Level

Total Record: 3658

Daily Monitoring Record Data

U ID.	Name	Guardian Name	Gender	Age	Contact	Travel Type	Travel Address	Travel Date	District	Block	Panchayat	Date	Slot	Officer Name	Officer Designation	Officer Contact	Officer Remark	Login User	Symptom	Quarantine	Available	Poster	Addr Link
3132	SHIVNATH SINGH	LT SNAJEEVAN SINGH	Male	52	7979842530	Domestic	KOLKATA	24-Mar-2020	Vaishali	Bidupur	Khanpur Pakri	02-Apr-2020	Evening	Bibha kumari	A F	8757477586	Good	Field Exec 721	None	at_home1	Yes	No	NA
3164	Sharmila devi	Awadesh paswan	Female	26	9801470758	Domestic	Delhi	14-Mar-2020	Vaishali	Bidupur	Khanpur Pakri	02-Apr-2020	Morning	Bibha kumari	Aasha	8757477586	Good	Field Exec 721	None	at_home1	Yes	No	NA
3219	Ganesh paswan	Rampravesh paswan	Male	26	9801470758	Domestic	Delhi	14-Mar-2020	Vaishali	Bidupur	Khanpur Pakri	02-Apr-2020	Morning	Bibha kumari	Aasha	8757477586	Good	Field Exec 721	None	at_home1	Yes	No	NA
3330	Raj kapur kumar	Sakaldeep ray	Male	25	7319900274	Domestic	Mumbai	23-Mar-2020	Vaishali	Bidupur	Sahdullahpur Dhasbauli	02-Apr-2020	Morning	ANIL GANMURTI	GAS	9334757224	Ok	Field Exec 721	None	at_home1	Yes	No	NA
3376	Mithun kumar	Sakaldeep ray	Male	22	7319900274	Domestic	Mumbai	23-Mar-2020	Vaishali	Bidupur	Sahdullahpur Dhasbauli	02-Apr-2020	Morning	ANIL GANMURTI	GAS	9334757224	Ok	Field Exec 721	None	at_home1	Yes	No	NA
3539	Mukesh singh	Dewan singh	Male	32	6203069326	Domestic	Punjab	23-Mar-2020	Vaishali	Bidupur	Khanpur Pakri	02-Apr-2020	Evening	Bibha kumari	A F	8757447586	Good	Field Exec 721	None	at_home1	Yes	No	NA
3554	RAVI KUMAR	SHIVNATH SINGH	Male	25	7979842530	Domestic	KOLKATA	24-Mar-2020	Vaishali	Bidupur	Khanpur Pakri	02-Apr-2020	Evening	Bibha kumari	A F	8757477586	Good	Field Exec 721	None	at_home1	Yes	No	NA
3607	Lalit singh	Uttamchandra singh	Male	40	8789905117	Domestic	Punjab	19-Mar-	Vaishali	Bidupur	Khanpur Pakri	02-Apr-	Evening	Bibha kumari	AF	8757477586	Good	Field Exec	None	at_home1	Yes	No	NA